SIYAMTHANDA MAFUTA

Computer Engineering Graduate | Software, Networks & Intelligent Systems

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PROFESSIONAL SUMMARY

Computer Engineering student with a completed Diploma in Computer Engineering and hands-on academic and project experience across software development, networking, databases, embedded systems and artificial intelligence. Currently developing practical solutions through university projects and the CAPACITI Demand Academy. Interested in graduate, internship and entry-level opportunities where I can contribute, learn and grow across software, IT infrastructure, networking and intelligent systems.

TECHNICAL SKILLS

- **Software & Development:** Python, Java, JavaScript, Node.js, HTML, CSS, Backend Development
- **Networks & Systems:** TCP/IP, Routing & Switching, Cisco Networking, Linux, Network Troubleshooting
- **AI & Data:** Artificial Intelligence, Machine Learning Fundamentals, Generative AI, Prompt Engineering, Signal Processing
- **Databases & Engineering:** MySQL, MongoDB, Database Design, ESP32, Arduino, Embedded Systems, MATLAB, Git/GitHub

PROFESSIONAL DEVELOPMENT

Demand Academy Participant | CAPACITI | Cape Town **2026 - Present**

- Participating in an intensive technology development programme focused on artificial intelligence, machine learning, software development and practical project work.
- Completed learning modules and certifications covering AI fundamentals, generative AI, Python for data science, prompt engineering and machine learning concepts.
- Collaborating on bootcamp projects and building a professional portfolio to showcase technical growth and project experience.

SELECTED PROJECTS

CSI-Heart - Contactless Vital Sign Extraction via Wi-Fi Sensing

Academic / Research Group Project | Project Team Member / Developer

Contributing to a system that explores Wi-Fi Channel State Information (CSI) for contactless vital-sign extraction using ESP32 devices, signal processing and machine-learning approaches.

RoadReady Traffic Services

CAPACITI / Demand Academy Collaborative Project | Project Team Member / Developer

Collaborative software project focused on digital traffic service requests and related operational workflows.

RoadReady Insights

CAPACITI / Demand Academy Collaborative Project | Project Team Member / Developer

Interactive platform for sharing traffic service experiences and insights.

InfraTrack System

Academic Group Project | Project Team Member / Developer

A full-stack ticket and infrastructure maintenance management system modelled around the City of Johannesburg Public Works Department workflow (academic prototype).

Duplicate File Finder

Personal Software Project | Individual Developer

Java-based application developed to identify and manage duplicate files.

EDUCATION

Bachelor of Engineering Technology in Computer Engineering

Cape Peninsula University of Technology (CPUT) | 2025 - 2026 (Currently Completing)

Diploma in Computer Engineering

Cape Peninsula University of Technology (CPUT) | 2020 - 2024 (Completed)

SELECTED CERTIFICATIONS

- Artificial Intelligence Bootcamp (AI) - CAPACITI / Coursera
- Google AI Essentials Specialization - Google
- Machine Learning: Regression and Classification - DeepLearning.AI
- Generative AI with Large Language Models - DeepLearning.AI / AWS
- Python for Data Science, AI & Development - IBM
- Introduction to Artificial Intelligence (AI) - IBM
- Additional CAPACITI learning: AI For Everyone, Prompt Engineering, Responsible AI and AI productivity courses